Practice Questions

MTH633

Lectures (23 – 25) or Modules (90 – 101)

Q1: What is Factor or quotient group? Give an example.

Q2: State and explain first isomorphism theorem.

Q3: Let (*G,* .) be a group with subgroup *H*. For $a, b \in G$, what do you mean by *a* congruent to *b* modulo

H?

Q4: Write down Two normal subgroups of $\{ 1, -1, i, -i, j, -j, k, -k \}$.

Q5: Define kernel of homomorphism, and give an example.

Q6: Let *y* be the natural map of *Z* into Z_n given by *y(m) = r*, where r is the remainder given by the division algorithm when *m* is divided by *n.* Show that *y* is a homomorphism. Find *Ker(y)*.